RAMDEOBABA

Idea Lab

24CS03PR0504

PROJECT TITLE

Multimodal AI-Based Machine Health Monitoring & Predictive Maintenance

- Vaishnav Balpande - B1-01
- Shreyank Nipane - B4-55
- Tanushri Sapate - B3-44
- Aastha Pusdekar- B4-52
- Kartik Madankar-B3-38
- Prof.Ashwini Lokhande

- Artificial Intelligence and Cyber Security
- 2026-27

PROBLEM STATEMENT

What problem are we addressing?

Industrial machines can develop faults through changes in temperature, vibration and acoustic behavior before visible or catastrophic failure. Existing monitoring systems often analyze these signals independently, leaving operators with fragmented information instead of a unified machine-health picture.



Who is affected?

- Industrial operators — need continuous visibility of machine condition.
- Maintenance teams — need earlier indication of developing abnormalities.
- Plant managers — face unplanned downtime and maintenance costs.
- Equipment owners — need scalable monitoring across multiple machines.

Current Challenges

- Single-sensor monitoring provides only partial information about machine condition.
- Raw vibration, acoustic and thermal data require specialized signal-processing expertise.
- Different fault indicators may appear across different sensing modalities.
- Conventional systems often provide separate measurements and alarms rather than unified evidence.

Why is the problem important?

Machine failures can lead to unplanned downtime, production loss, maintenance costs and equipment damage. A system capable of continuously combining multiple physical indicators can provide a more comprehensive basis for early anomaly detection and predictive maintenance.

EXISTING SOLUTION / LITERATURE REVIEW

Vibration-Based Monitoring

- Uses accelerometers to monitor rotating-machine vibration.
- Common analysis: RMS, Kurtosis, Crest Factor, FFT and spectral analysis.
- Effective for detecting mechanical and bearing-related abnormalities.
- Limitation: Primarily represents mechanical behavior and may miss thermal or acoustic indicators.

Acoustic / Sound Monitoring

- Uses microphones to capture machine acoustic signatures.
- Analysis includes FFT, STFT, Octave / ⅓-Octave and MFCC.
- Can identify abnormal sound patterns associated with machine operation.
- Limitation: Sensitive to environmental noise, operating conditions and recording location.

Thermal / Infrared Monitoring

- Uses thermal imaging to identify hotspots, thermal gradients and abnormal heating.
- Useful for observing friction, overheating and temperature-pattern changes.
- Provides spatial information unavailable from point sensors.
- Limitation: Thermal inspection alone cannot directly characterize vibration or acoustic faults.

Conventional Single-Modality Systems

- Typically monitor one sensor type independently.
- Generate separate alarms, measurements or inspection reports.
- Require operators to interpret multiple sources manually.
- Limitation: No unified interpretation of thermal + vibration + acoustic evidence.
- Bottom comparison bar

Existing approaches analyze machine condition mainly through individual sensor modalities. IDEA LAB combines thermal, vibration and acoustic evidence with signal processing, AI-based inference and confidence-aware fusion to provide a unified machine-health view.

PROPOSED SOLUTION

A real time system that combines vibration, acoustic and thermal data to monitor machine health more accurately.

Machine Sensors → **Process & Analyse** → **Combine the Evidence** → **Display to User**



Vibration

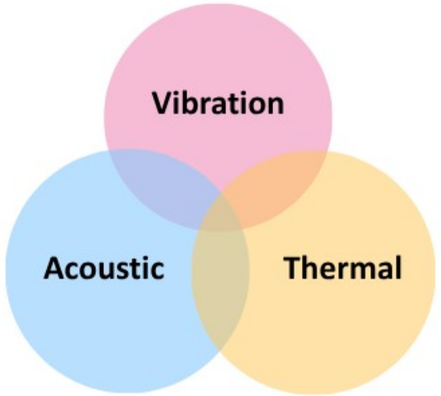


Acoustic

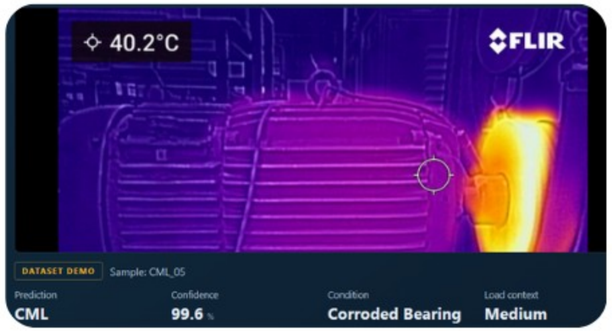


Thermal

- Preprocess data (remove noise , clean)
- Extract useful features
- Detect unusual patterns



Get overall machine heath



- Instead of relying on a single sensor , our system combines vibrations, acoustic and thermal information to get a **more complete and reliable picture of machine health.**

OBJECTIVES



Acquire & Process Data

Collect and process thermal, vibration, and acoustic machine data while extracting relevant features.



Detect anomalies

Analyze vibration, acoustic, and thermal patterns to identify abnormal machine behaviour.



Fuse Evidence

Combine multimodal sensor results to generate an overall machine-health assessment with confidence.



Build Monitoring Dashboard

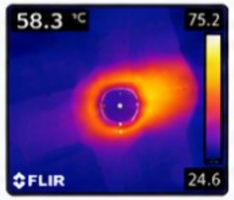
Display live telemetry, machine health,trends, events, alerts, and system status in a unified interface.

Measurable System Goals

- Real-time telemetry and health updates.
- Sensor analysis with missing-data handling.
- Event and history visualization.
- Support for demo and live data.

SYSTEM ARCHITECTURE / METHODOLOGY

1 INPUT / DATA ACQUISITION
Multimodal Sensing



THERMAL
- Thermal Camera / Dataset
- Frame-by-frame thermal imagery



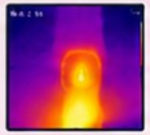
VIBRATION
- Vibration Sensor / IMS Dataset
- Windowed signals (20 kHz, 20,480 samples)



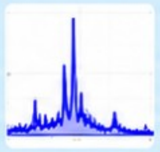
ACOUSTIC
- Microphone / Acoustic Dataset
- Audio recordings (16 kHz, 10 s)

Machine → Sensors / Dataset → Edge Processing

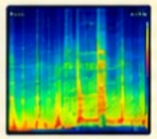
2 SIGNAL PROCESSING & FEATURE EXTRACTION
Modality-specific analysis



THERMAL
- Frame preprocessing
- Relative intensity
- Hotspot area & centroid
- Hotspot gradients
- Hotspot evolution



VIBRATION
- Sliding-window analysis
- RMS, Kurtosis, Crest Factor
- FFT & spectral features
- Spectral centroid/bandwidth
- Octave / 1/3-Octave energy



ACOUSTIC
- Audio preprocessing
- FFT & STFT
- Octave / 1/3-Octave bands
- MFCC
- Spectral features

Output:
Cleaned signals + modality-specific health features

3 AI / ANOMALY INTELLIGENCE
Deterministic + Machine Learning

Deterministic Health Engine
- Baseline calibration
- Statistical z-score detection
- Isolation Forest
- Severity estimation
- Temporal smoothing / hysteresis

Modality-specific ML Models

Thermal	→	MobileNetV3-Small (Transfer Learning)
Vibration	→	Random Forest
Acoustic	→	Logistic Regression

Confidence Gating
- Suppress low-confidence predictions
- Reliability score per modality

Multimodal Evidence Fusion
- Confidence-weighted fusion
- Handle missing modalities
- ML evidence is supplementary; deterministic health remains authoritative.

4 OUTPUT / MONITORING
Actionable Machine-Health View

Machine Health Score



Current State

Normal | Warning | Critical | Unknown

Multimodal Evidence
- Thermal – prediction + confidence
- Vibration – prediction + confidence
- Acoustic – prediction + confidence
- Fused ML – severity + confidence

Events & History
- Anomaly events
- Health trends
- Historical telemetry

Operator Interface
Real-time monitoring dashboard (React + TypeScript)

HARDWARE / DATA SOURCES
Thermal Camera | Vibration Sensor | Microphone | Edge Computer
(Demo: Dataset-driven thermal / vibration / acoustic sources)

SOFTWARE / TECHNOLOGIES
Python | FastAPI | PyTorch | scikit-learn | NumPy | SciPy | librosa | OpenCV | SQLite | React | TypeScript | WebSocket

IMPLEMENTATION

Thermal Camera
Thermal imagery & hotspot monitoring



Vibration Sensor
Rotating-machine vibration acquisition

Microphone
Machine acoustic signatures

Edge Computer
Local processing, inference & fusion

Prototype Environment
Prerecorded datasets currently simulate real sensor inputs



Technologies Used

Backend & Real-time
- Python + FastAPI
- SQLite + WebSocket

Monitoring Dashboard
- React + TypeScript + Vite

Machine Learning
- PyTorch (Thermal)
- scikit-learn (Vib & Acou)

Signal Processing
- NumPy
- SciPy
- librosa

System Implementation

Multimodal Pipeline
- Real-time processing
- Thermal frame analysis
- Vibration window analysis
- Audio window analysis

Health Engine
- Automatic baseline calibration
- Deterministic machine-health engine
- Temporal smoothing & hysteresis
- Missing-modality handling

Telemetry & Persistence
- Live WebSocket telemetry
- Event & history persistence

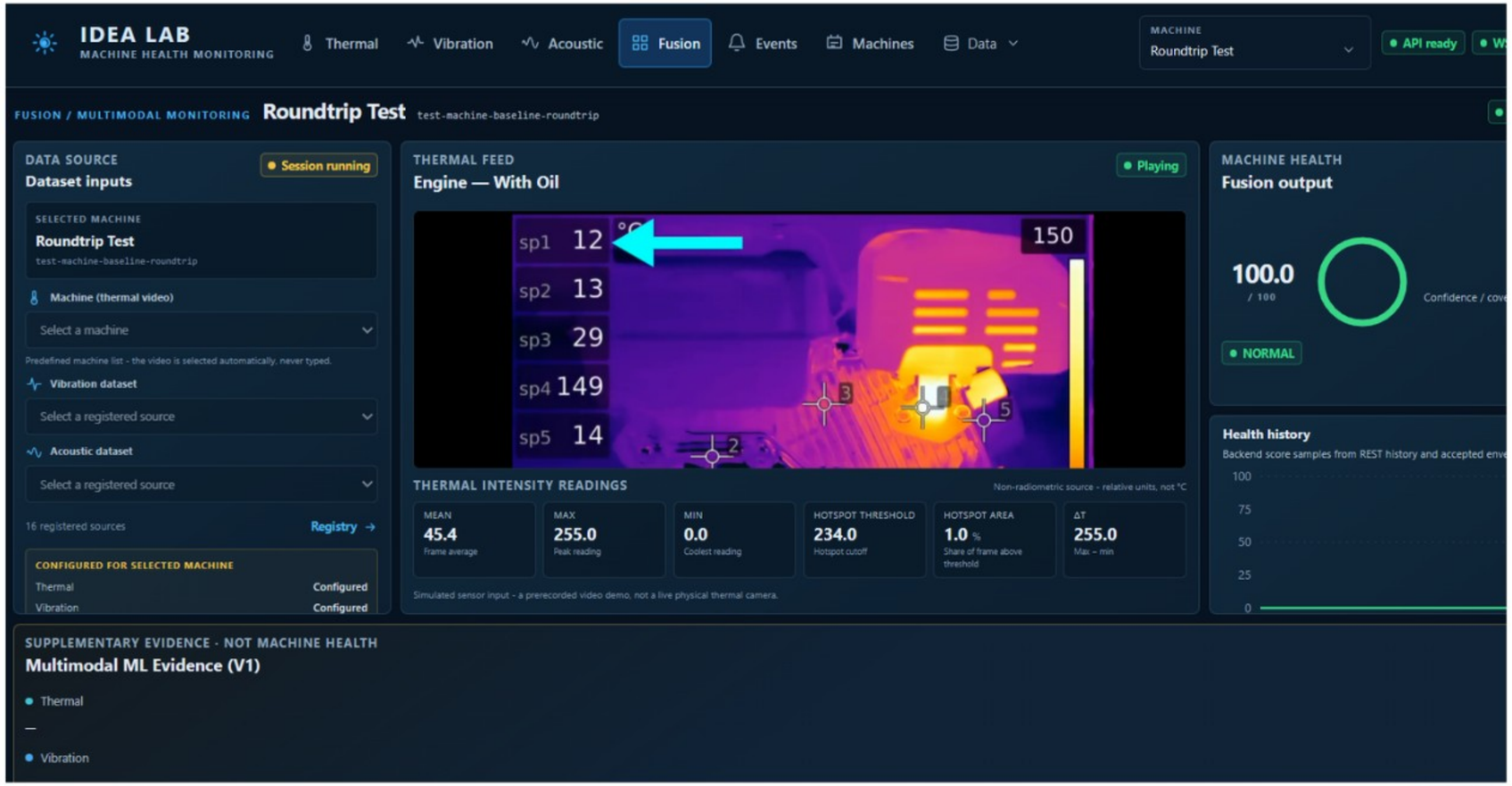
AI & Signal Processing

ML Models
- Thermal: MobileNetV3-Small (Transfer Learning)
- Vibration: Random Forest
- Acoustic: Logistic Regression (61 features)

Analysis & Features
- FFT / STFT
- Octave / ⅓-Octave
- MFCC

Anomaly & Fusion
- Statistical + Isolation Forest
- Confidence Gating
- Weighted ML Evidence Fusion

RESULTS / PROTOTYPE



CAPABLITIES

- Thermal video-based monitoring
- Vibration signal analysis
- Acoustic signal analysis
- Modality-specific ML inference
- Confidence-gated predictions
- Multimodal evidence fusion
- Real-time health dashboard
- Event & health-history tracking

KEY RESULTS

3 Sensing Modalities
Thermal • Vibration • Acoustic

3 ML Models
CNN • Random Forest • Logistic Regression

182+ Backend Tests Passed
Unit • Integration • API

DEMO MACHINE SET

7 Usable Videos | **6** Machine Types



Honda Engine (V1, V2) | Hydraulic System | EV Battery | Transformer | Engine With Oil | Engine Without Oil

INNOVATION, IMPACT, DELIVERABLES & FUTURE SCOPE

1 - INNOVATION



- **Modality Intelligence:** Preserves domain signal features before ML.
- **Evidence-Level Fusion:** Combines normalized evidence without forcing unaligned raw feeds.
- **Confidence Gating:** Suppresses uncertain model inferences to avoid alert fatigue.

2 - IMPACT



- **Downtime Prevention:** Detects multi-sensor precursors before breakdown.
- **Lower Skill Barrier:** Non-experts read a simple 0–100% composite health index.
- **Edge Autonomy:** Eliminates cloud dependency, ensuring real-time response.

3 - DELIVERABLES



- **Backend Server:** FastAPI + WebSocket real-time analytics engine.
- **Operator Cockpit:** React + Vite dashboard with live spectrum views.
- **AI Pipeline:** Edge-optimized MobileNetV3 + Isolation Forest modules.

4 - FUTURE



- **Phases:** Specialized vibration and acoustic deep learning models.
- **RUL Estimation:** Remaining Useful Life regression based on trajectory.
- **Plant Pilot:** Direct hardware DAQ interfacing with shop-floor motors.

CONCLUSION & REFERENCES

CONCLUSION

Our Solution is a multimodal machine-health monitoring platform that combines thermal, vibration and acoustic intelligence in a single edge-oriented system. Each modality is independently processed using domain-specific signal analysis and machine learning, followed by confidence-aware evidence fusion and deterministic health assessment. The integrated system provides continuous machine-health status, anomaly detection, event tracking and real-time visualization through a unified dashboard.

KEY OUTCOME

- Three sensing modalities: Thermal + Vibration + Acoustic
- Three modality-specific ML models: MobileNetV3-Small + Random Forest + Logistic Regression
- Confidence-gated predictions to suppress uncertain ML outputs
- Multimodal weighted evidence fusion with missing-modality handling

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